1st Midterm · Unit 1 · First-order modelling · 5 points · 30 min · Standard exam level

P01 — Two brine tanks in cascade: separation for the first tank, integrating factor for the second, the equal-rate case and the peak of salt

Two well-stirred tanks are connected in series. Tank A holds $V_A = 100$ L of brine containing 20 kg of dissolved salt; tank B holds V_B litres of fresh water. At $t = 0$ three pumps are switched on, all with the same flow rate $r = 10$ L/min: fresh water enters A, the mixture of A flows into B, and the mixture of B is drained to the outside. Time is measured in minutes. Let $x(t)$ and $y(t)$ be the masses of salt (kg) in A and in B, and write $k_A = r/V_A$, $k_B = r/V_B$ for the rate constants. Part (c) uses $V_B = 100$ L and part (d) uses $V_B = 50$ L.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns $x(t), y(t)$ (kg), independent variable $t \geq 0$ (min).
- Hypotheses: perfect mixing (the concentration of the outflow of each tank is its mean concentration), equal inflow and outflow, so the volumes stay constant; salt is conserved (no reaction, no precipitation).
- Operator: two first-order, linear equations with constant coefficients. The equation for x is homogeneous and decoupled; the equation for y is inhomogeneous, forced by x (a triangular, one-way coupling).

Quantity	Symbol	Value	Unit
Flow rate	r	10	L/min
Volume of A	V_A	100	L
Volume of B	V_B	100 in (c), 50 in (d)	L
Initial salt in A	x_0	20	kg
Initial salt in B	$y(0)$	0	kg
Rate constants	$k_A = r/V_A$, $k_B = r/V_B$	0.1; 0.1 or 0.2	min^{-1}

Phase 2. Canonical formulation

The balance law (rate of change = rate in − rate out) for each tank, in the linear standard form $y' + p(t)\,y = q(t)$ of Topic 2:

$$x' + k_A\,x = 0, \qquad y' + k_B\,y = k_A\,x(t), \qquad x(0) = x_0, \quad y(0) = 0.$$

Part (a) · 1 point

Model and dimensions

Write the initial value problem satisfied by x and y and check its dimensions.

Step 1. Tank A

Fresh water carries no salt, so the rate in is 0. The outflow carries r L/min at concentration x/V_A kg/L:

$$\frac{dx}{dt} = 0 - r\,\frac{x}{V_A} = -k_A\,x, \qquad k_A = \frac{r}{V_A}.$$

Step 2. Tank B

Its inflow is the outflow of A, with concentration x/V_A; its outflow has concentration y/V_B:

$$\frac{dy}{dt} = r\,\frac{x}{V_A} - r\,\frac{y}{V_B} = k_A\,x - k_B\,y, \qquad k_B = \frac{r}{V_B}.$$

The initial data are $x(0) = x_0 = 20$ kg and $y(0) = 0$ (fresh water).

Step 3. Dimensions

$[r] = \text{L min}^{-1}$, $[V] = \text{L}$, so $[k_A] = [k_B] = \text{min}^{-1}$. Then $[k_A x] = \text{min}^{-1}\cdot\text{kg} = \text{kg/min} = [dx/dt]$, and the same for every term of the second equation: both equations are dimensionally homogeneous. The exponents that will appear, $k_A t$ and $k_B t$, are dimensionless.

Numerically $k_A = 10/100 = 0.1$ min^{-1}; $k_B = 10/100 = 0.1$ min^{-1} for $V_B = 100$ L and $k_B = 10/50 = 0.2$ min^{-1} for $V_B = 50$ L.

KEY RESULTS

- $x' = -\dfrac{r}{V_A}x,\ y' = \dfrac{r}{V_A}x - \dfrac{r}{V_B}y,\ x(0) = 20$ kg, $y(0) = 0$.
- $k_A = k_B = 0.1$ min^{-1} in (c); $k_A = 0.1$, $k_B = 0.2$ min^{-1} in (d).
- Every term has units kg/min.

Part (b) · 1.5 points

Solution for different rate constants

Solve for $x(t)$ by separation of variables and, for $k_A \neq k_B$, solve for $y(t)$ with an integrating factor.

Step 1. Tank A by separation

$x \equiv 0$ is not the solution ($x_0 \neq 0$). For $x > 0$, separate: $\dfrac{dx}{x} = -k_A\,dt$. Integrate from 0 to t:

$$\int_{x_0}^{x(t)} \frac{d\xi}{\xi} = \big[\ln \xi\big]_{x_0}^{x(t)} = \ln \frac{x(t)}{x_0}, \qquad \int_0^t (-k_A)\,ds = -k_A\,t.$$

Hence $\ln(x/x_0) = -k_A t$ and

$$x(t) = x_0\, e^{-k_A t}.$$

Step 2. Integrating factor for B

Substitute $x(t)$: $y' + k_B y = k_A x_0 e^{-k_A t}$. With $p = k_B$ the integrating factor (Notes, p. 3) is $\mu(t) = e^{\int_0^t k_B\,ds} = e^{k_B t}$, with $\mu' = k_B \mu$. By the product rule, $(\mu y)' = \mu y' + \mu' y = e^{k_B t}(y' + k_B y)$, so

$$\frac{d}{dt}\big(e^{k_B t} y\big) = k_A x_0\, e^{(k_B - k_A)t}.$$

Step 3. Integration

For $k_B \neq k_A$ an antiderivative of $e^{(k_B-k_A)s}$ is $\dfrac{e^{(k_B-k_A)s}}{k_B - k_A}$. Integrating from 0 to t and using $y(0) = 0$ (Barrow's rule on both sides):

$$e^{k_B t} y(t) - e^0\, y(0) = k_A x_0 \left[\frac{e^{(k_B-k_A)s}}{k_B - k_A}\right]_0^t = \frac{k_A x_0}{k_B - k_A}\Big(e^{(k_B-k_A)t} - 1\Big).$$

Multiplying by $e^{-k_B t}$:

$$y(t) = \frac{k_A\, x_0}{k_B - k_A}\Big(e^{-k_A t} - e^{-k_B t}\Big).$$

Step 4. Check

$y(0) = \frac{k_A x_0}{k_B - k_A}(1 - 1) = 0$. Differentiating, $y' = \frac{k_A x_0}{k_B - k_A}(-k_A e^{-k_A t} + k_B e^{-k_B t})$ and $k_A x - k_B y = \frac{k_A x_0}{k_B - k_A}\left[(k_B - k_A)e^{-k_A t} - k_B e^{-k_A t} + k_B e^{-k_B t}\right]$, which is the same expression. The quotient $\frac{e^{-k_A t} - e^{-k_B t}}{k_B - k_A}$ does not change when k_A and k_B are exchanged (numerator and denominator both change sign), and it is positive for $t > 0$: $y(t) > 0$ whichever tank is faster.

KEY RESULTS

- $x(t) = x_0\, e^{-k_A t}$.
- $y(t) = \dfrac{k_A\, x_0}{k_B - k_A}\big(e^{-k_A t} - e^{-k_B t}\big)$, $k_A \neq k_B$.

Part (c) · 1.5 points

Equal volumes: the resonant case

Take $V_B = 100$ L. Solve the equation for y directly, show that the result is the limit of (b) as $k_B \to k_A$, and find the time and value of the largest amount of salt in B.

Step 1. Direct solution

With $k_A = k_B = k = 0.1\ \text{min}^{-1}$, Step 2 of (b) gives $\frac{d}{dt}(e^{kt}y) = kx_0 e^0 = kx_0$. Integrating from 0 to t: $e^{kt}y(t) - y(0) = kx_0\,[s]_0^t = kx_0\, t$, so

$$y(t) = k\, x_0\, t\, e^{-kt} = 0.1 \cdot 20\, t\, e^{-t/10} = 2\, t\, e^{-t/10}\ \text{kg}.$$

The forcing $kx_0 e^{-kt}$ decays at exactly the rate of the free response e^{-kt} of tank B: the exponential guess fails and a factor t appears, the first-order analogue of resonance.

Step 2. Limit of the general formula

Fix $k_A = k$. In $y = kx_0\,\dfrac{e^{-kt} - e^{-k_B t}}{k_B - k}$ numerator and denominator vanish at $k_B = k$. By L'Hôpital's rule in the variable k_B, with $\frac{\partial}{\partial k_B} e^{-k_B t} = -t\, e^{-k_B t}$ (chain rule):

$$\lim_{k_B \to k} \frac{e^{-kt} - e^{-k_B t}}{k_B - k} = \lim_{k_B \to k} \frac{t\, e^{-k_B t}}{1} = t\, e^{-kt},$$

so $y \to k x_0\, t\, e^{-kt}$, the result of Step 1. The two cases join continuously.

Step 3. Maximum

By the product rule, $y'(t) = k x_0\big(e^{-kt} + t(-k)e^{-kt}\big) = k x_0\, e^{-kt}(1 - kt)$. It is positive for $t < 1/k$ and negative for $t > 1/k$, so y has its maximum at $t^* = 1/k = 10$ min:

$$y_{\max} = k x_0 \cdot \frac{1}{k}\, e^{-1} = \frac{x_0}{e} = \frac{20}{e} \approx 7.358 \text{ kg},$$

a concentration of $7.358/100 \approx 0.0736$ kg/L. At that moment tank A holds $x(10) = 20e^{-1}$ kg as well.

KEY RESULTS

- $y(t) = k\, x_0\, t\, e^{-kt} = 2\, t\, e^{-t/10}$ kg.
- Maximum at $t^* = 1/k = 10$ min, $y_{\max} = x_0/e = 20/e \approx 7.358$ kg (concentration 0.0736 kg/L).

Part (d) · 1 point

Smaller second tank, peak and long-time behaviour

Take $V_B = 50$ L. Find the time and value of the largest amount of salt in B, compare the concentrations of both tanks then, give the limit $t \to \infty$ and the total salt that leaves the system.

Step 1. The formula

$k_A = 0.1, k_B = 0.2$ min^{-1}, so $\frac{k_A x_0}{k_B - k_A} = \frac{0.1 \cdot 20}{0.1} = 20$ kg and $y(t) = 20\big(e^{-t/10} - e^{-t/5}\big)$.

Step 2. Maximum

$y'(t) = 20\big(-\frac{1}{10}e^{-t/10} + \frac{1}{5}e^{-t/5}\big)$. With $u = e^{-t/10} \in (0, 1]$: $y' = 20\big(-\frac{u}{10} + \frac{u^2}{5}\big) = \frac{20u}{10}(2u - 1)$, which vanishes for $u = \frac{1}{2}$, is positive for $u > \frac{1}{2}$ (early times) and negative after. Hence $e^{-t^*/10} = \frac{1}{2}$, $t^* = 10 \ln 2 \approx 6.93$ min, and

$$y_{\max} = 20\Big(\frac{1}{2} - \frac{1}{4}\Big) = 5 \text{ kg}.$$

This agrees with the general time $t^* = \dfrac{\ln(k_B/k_A)}{k_B - k_A} = \dfrac{\ln 2}{0.1}$.

Step 3. Concentrations at the peak

$x(t^*) = 20 \cdot \frac{1}{2} = 10$ kg, so A has $10/100 = 0.1$ kg/L and B has $5/50 = 0.1$ kg/L. This is general: $y' = 0$ means $k_A x = k_B y$, i.e. $\frac{x}{V_A} = \frac{y}{V_B}$. Tank B stops gaining salt exactly when its concentration equals that of the brine coming in.

Step 4. Long time and total outflow

Both exponentials tend to 0, so $x(t) \to 0$ and $y(t) \to 0$: all salt is washed out. The salt drained from B per minute is $k_B y$; with the antiderivatives $-10e^{-t/10}$ and $-5e^{-t/5}$ and Barrow's rule on $[0, T]$, $T \to \infty$:

$$\int_0^{\infty} 0.2 \cdot 20\big(e^{-t/10} - e^{-t/5}\big)\,dt = 4\Big[-10e^{-t/10} + 5e^{-t/5}\Big]_0^{\infty} = 4\big(0 - (-10 + 5)\big) = 20 \text{ kg},$$

the whole initial mass x_0, as conservation of salt requires.

KEY RESULTS

- $y(t) = 20\big(e^{-t/10} - e^{-t/5}\big)$ kg.
- $t^* = 10 \ln 2 \approx 6.93$ min, $y_{\max} = 5$ kg; at t^* both concentrations equal 0.1 kg/L.
- $x, y \to 0$; total outflow of salt $\int_0^{\infty} k_B y\,dt = 20$ kg.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to 0^+$: $y \approx k_A x_0\, t$ (B starts filling at the rate A loses salt, $k_A x_0 = 2$ kg/min). $t \to \infty$: $y \sim \frac{k_A x_0}{k_B - k_A} e^{-k_A t}$ when $k_A < k_B$, so the slower tank sets the decay. $V_B \to 0$ ($k_B \to \infty$): $y \to 0$, B becomes a pipe.
- **Checks:** both formulas satisfy the ODEs and $y(0) = 0$ by substitution; the equal-rate case is the limit of the general one; $\int_0^{\infty} k_B y\,dt = x_0$. Dimensions: $k\,x_0\,t$ is $(\text{min}^{-1})(\text{kg})(\text{min}) = \text{kg}$, $t^* = 1/k$ and $\ln(k_B/k_A)/(k_B - k_A)$ are in minutes.
- **Common mistakes in the exam:** (1) using the volume of B in the inflow term of B (the inflow concentration is x/V_A); (2) applying the formula of (b) with $k_A = k_B$, which divides by zero, instead of noticing the factor t; (3) dropping the lower limit in the integrating-factor integral, which leaves $y(0) \neq 0$; (4) reporting the maximum time without checking the sign of y', or confusing the maximum of mass with the maximum of concentration.